

Luke Crimi

Computer Science and Engineering, Lehigh University - Class of 2027

Interested in robotics, embedded software engineering, and automation.

U.S. Citizen

Github: CodeSample15

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Florham Park, NJ

EDUCATION

Lehigh University

08/2024 - present (transfer student from Rutgers University)

- Major: Computer Science and Engineering
 - Expected graduation: Spring 2027
 - GPA: 3.81
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CERTIFICATION

- FAA Part 107 Drone License
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EXPERIENCE

Lutron Electronics — Embedded Software Engineering Summer Co-op

Summer 2026

- Successfully completed the implementation and documentation of a Go application that runs on Lutron embedded processors to diagnose network connectivity issues.

Lehigh University — Lab technician and Research assistant

- **Lab technician (Spring semester 2025):** Set up and configured robots using ROS for students in an introduction to mobile robots class. Wrote automation scripts in Python to expedite setup process.
- **Research assistant (2025 - present):**
 - Studying how artificial intelligence can assist students who need help with their classes.
 - Used Python to write web scraping scripts for data collection.
 - Created a Chrome extension using Typescript, React, Bootstrap, and a Cloudflare worker which enables students to use LLMs to assist them with coursework uploaded to Lehigh's Coursesite page. Worked with various popular APIs such as Google's OAuth and OpenAI's text completion.

Robot Revolution | Summit, NJ — Teacher

2020 - 2023

- Taught an intermediate C++ course. Helped to design the curriculum, and worked with the owner of Robot Revolution to decide what topics would be covered in class. Guided students to improve their programming skills.

Code Ninjas | Denville, NJ — Sensei

2022 - 2023

- Assisted students with the development of several small computer programs. Led presentations to students regarding learning topics. Completed various setup and cleanup tasks.

New Horizons Day Camp | Florham Park, NJ — Summer Counselor

2021 - 2024

- Ran the rocketry activity for grades 2-9. Worked with a team of counselors to prepare for and organize the camp's rocket launch event. Improved time management and interpersonal skills.

Ocean Beach Trading | Ocean Beach, NY — Retail Floater

06/2024 - 08/2025

PERSONAL PROJECTS

Scribble - C++

- My second attempt at making an esoteric language for game development.

Squiggly - C++, Linux, Raspberry Pi

- Esoteric programming language developed in C++ designed to create simple games using scripts
- Ported to a handheld console powered by a Raspberry Pi

Keras Code Generator - C#, Unity, app development, Keras

- Graphical interface for creating Keras models in Python. Compiles user-generated graph structures into Python code

Bluetooth LED Light Controller - Bluetooth, Arduino, Raspberry Pi, Python

- Created an Arduino LED light controller that communicates with a Raspberry Pi over Bluetooth
- Raspberry Pi runs custom light patterns on a LED strip written in Python

Touchless Computer Interaction (eye/face tracking) - Python, Artificial Intelligence (Mediapipe)

- Tracks the location and rotation of a user's face through a web camera to move a cursor across the screen. Uses facial landmarks to determine when the user winks to emulate a mouse click

Wand Controlled LED Lights - Python, Computer vision, Artificial Intelligence, Raspberry Pi, 3D Printing, Keras

- Uses a IR camera and a custom trained CNN model to track patterns made with a IR light reflecting wand tip

Personal Website - React, Typescript, Bootstrap

- Created a simple website to host pages for some of my personal projects.
 - <https://lukecrimi.com/projects>
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Proficient programming languages: C++/C, Python, Go